Technical Design Report — Cryogenic Heat Load Analysis

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GBOGEB/ABACUS Schema: v2.1.0 · **Governance:** `verification_hook` validated

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1. Introduction

This report presents the thermal design analysis and mass-balance verification for the **European Spallation Source — Cryogenic Distribution System**. The facility operates **23 superconducting cryomodules** at a nominal temperature of **2.0 K** using **Helium-4** as the primary coolant, below the superfluid lambda point (2.1768 K).

1.1 Scope

- Static and dynamic heat load budget
- Coolant mass-flow rate verification
- Process topology review (cryomodule → valve → heat exchanger)
- GBOGEB/ABACUS governance integration

1.2 References

ID	Reference
[1]	NIST Cryogenics Technologies Group, REFPROP v10
[2]	Donnelly & Barenghi, J. Phys. Chem. Ref. Data **27**, 1217 (1998)
[3]	GBOGEB/ABACUS Governance Engine Specification, v6.0

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2. Cryogenic Operating Conditions

The helium circuit operates in a **sub-lambda** regime (He-II, superfluid), providing exceptional thermal conductivity for SRF cavity cooling.

Parameter	Value	Unit
Supply Temperature	2.0	K
Return Temperature	4.5	K
Supply Pressure	1.2	bar
Return Pressure	1.05	bar
Lambda Point	2.1768	K
Supply Enthalpy	2.56	J/g
Return Enthalpy	26.14	J/g
Enthalpy Difference (Δh)	23.58	J/g

2.1 Phase Regime

```
T_supply = 2.0 K < λ = 2.1768 K  →  He-II (superfluid) ✓
T_return = 4.5 K > λ = 2.1768 K  →  He-I (normal fluid)
```

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3. Heat Load Budget

3.1 Category Breakdown

Category	ID	Value (W)	Tolerance (%)	Type
Static — Radiation	`static_radiation`	12.5	±5	Static
Static — Conduction	`static_conduction`	8.3	±8	Static
Static — Supports	`static_supports`	3.7	±10	Static
Dynamic — RF losses	`dynamic_rf`	28.0	±3	Dynamic
Dynamic — Beam-induced	`dynamic_beam`	15.6	±6	Dynamic
Dynamic — HOM power	`dynamic_hom`	4.2	±12	Dynamic

3.2 Summary

Metric	Symbol	Value
Static heat load	Q_static	**24.5 W**
Dynamic heat load	Q_dynamic	**47.8 W**
Total heat load	Q_total	**72.3 W**
Safety factor	SF	1.5
Design heat load	Q_design	**108.45 W**

3.3 Distribution Visualization

```
 Static (34%)  ████████████▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒  24.5 W
Dynamic (66%)  ████████████████████████▒▒▒▒▒▒▒▒▒▒  47.8 W
──────────────────────────────────────────────────────────
Total          ████████████████████████████████  72.3 W
```

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4. Mass-Balance Verification

4.1 Governing Equation

The required coolant mass-flow rate is derived from the first law of thermodynamics:

$$ \dot{m} = \frac{Q_{\text{static}} + Q_{\text{dynamic}}}{\Delta h} $$

where $\Delta h = h_{\text{return}} - h_{\text{supply}}$.

4.2 Calculation

```
 # Mass balance calculation
Q_static  = 24.5   # W
Q_dynamic = 47.8   # W
Q_total   = 72.3   # W

h_supply  = 2.56   # J/g  (He-II @ 2.0 K, 1.2 bar)
h_return  = 26.14  # J/g  (He-I  @ 4.5 K, 1.05 bar)
delta_h   = 23.58  # J/g
```

```
m_dot_calc   = Q_total / delta_h      # = 3.066 g/s
m_dot_design = m_dot_calc * 1.5       # = 4.599 g/s (SF=1.5)
```

4.3 Verification Result

Parameter	Calculated	Design	Status
ṁ (g/s)	3.066	4.600	✓ PASS
Deviation	—	< 1%	✓ PASS

Result: PASS — Mass-flow rate within design envelope.

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5. Process Topology

5.1 Flowchart

```
graph LR
    A[CM-01<br/>Cryomodule<br/>T=2.01 K] -->|LHe supply| B[CV-01<br/>Control Valve<br/>45%]
    B -->|GHe return| C[HX-01<br/>Heat Exchanger<br/>ε=94.2%]
    style A fill:#e3f2fd,stroke:#1a73e8,stroke-width:2px
    style B fill:#fff3e0,stroke:#ea8600,stroke-width:2px
    style C fill:#e8f5e9,stroke:#34a853,stroke-width:2px
```

5.2 Component Parameters

Component	Type	Key Parameter	Status
CM-01	Cryomodule	T = 2.01 K, Q = 72.3 W	Nominal
CV-01	Control Valve	Position = 45%, Cv = 12.8	Nominal
HX-01	Heat Exchanger	ε = 94.2%, Duty = 108.45 W	Nominal

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6. GBOGEB/ABACUS Governance Integration

This analysis integrates with the GBOGEB/ABACUS governance framework:

```
governance:
  verification_hook: "engines.verification_hook"
```

```
  semantic_theme_tokens:
    primary: "var(--color-primary)"
    accent: "var(--color-accent)"
  render_rules:
    figure_prefix: "fig"
    equation_numbering: true
    cross_ref_validation: true
```

6.1 Verification Hooks

Hook	Engine	Status
YAML schema validation	`verification_hook`	✓
Figure cross-references	`render_rules`	✓
Equation numbering	`render_rules`	✓
Theme token binding	`SEMANTIC_THEME`	✓

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7. Conclusions

1. **Heat load budget** is fully characterized with 6 categories totaling **72.3 W**
2. **Mass balance** verified: $\dot{m}$ = 3.066 g/s (design: 4.6 g/s with SF=1.5) — **PASS**
3. **He-II superfluid** regime confirmed at supply conditions (2.0 K < λ)
4. **Heat exchanger** effectiveness (94.2%) exceeds 90% minimum
5. **GBOGEB/ABACUS** governance hooks validated end-to-end

Recommendation: Proceed to detailed engineering phase.

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